

4CS015 – Workshop #5 TO BE SUBMITTED AS PART OF YOUR PORTFOLIO – please note this workshop covers material from lectures 3,4,5 and 6 (next week).

Name:

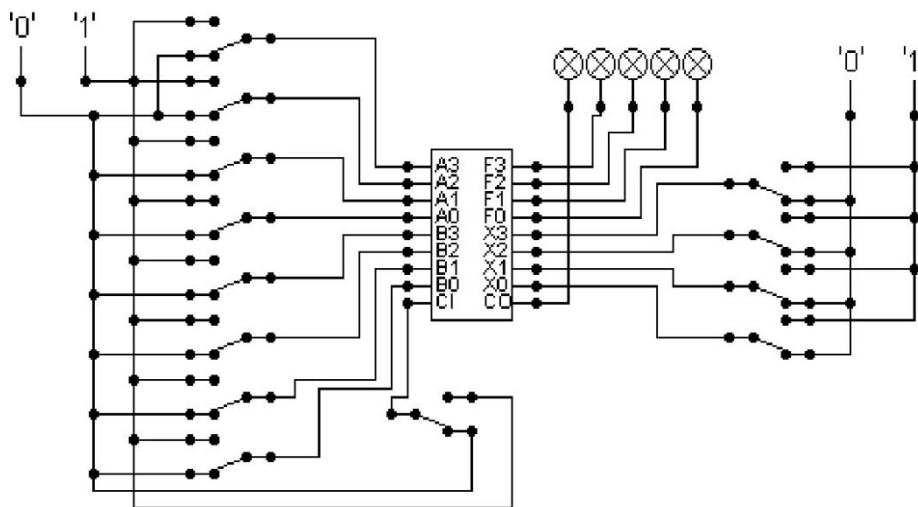
Student ID:

This is a marked workshop. It forms the second part of your portfolio. You will need to complete the workshop and then submit a copy of this document with a title that follows the following format (“DENNETT 1234567 wsp5.docx”), via CANVAS, by the deadline.

Workshop tasks:

Arithmetic Logic Unit

Load the LogSim Arithmetic Logic Unit Circuit **alu.cct** from inside the logsim application (You'll find it in the logsim folder) (***You may need to right-click on the link to download the file instead of opening it in the browser***). It should look like this:



The circuit behaves like a simple arithmetic logic unit. The inputs A0-A3 represent a 4 bit binary number. Inputs B0-B3 represent another binary number. A0 and B0 are the least significant bits respectively. The following table details the functions supported by the chip. All other control lines = 0.

Function	AND	OR	XOR	NAND	NOR	NOT A	ADD	SUBTRACT
X3 – X0	0000	0001	0010	0011	0100	0101	1010	1011

Use A= 11 B=4, complete the following table in binary **(10 marks)**:

FUNCTION	OUTPUT
AND	00000
OR	01111
XOR	01111
NAND	01111
NOR	00000
NOT A	00100
ADD	01111
SUBTRACT	00111

The logical operations are bitwise. Manually prove each operation has returned the correct result by **(15 marks)**:

Example:

1 0 1 1

0 1 0 0 AND OPERATION

0 0 0 0 RESULT

MANUAL WORKINGS:

AND; = 00000

1011

0100

OR;= 01111

1011

0100

XOR; = 01111

1011

0100

NAND; = 01111

1011

0100

NOR; = 00000

1011

0100

NOT A; = 00100

1011

0100

ADD; =01111

1011

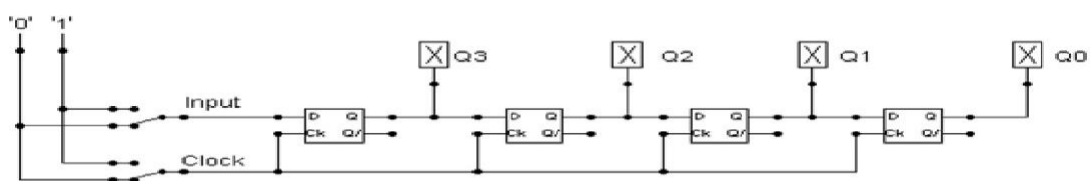
0100

SUBTRACT; = 00111

1011

0100

Lecture 6 material: Serial to Parallel Decoder:



Build the circuit above and complete the following timing diagram by filling in the table spaces with '1' or '0'. **(15 marks)**

Remember to disable preset / clear for the d-type in properties.

Work from left to right for the inputs, when you run your logic circuit get the output at **1111** with **10** input, this will be your **first input column** (remember your first gate is Q3 which is at the top of the table). You may need to clock 0-1 a few times when the circuit is running to get the output required. Note where Q3-Q0 are on the table and on the logic circuit.

Input	1	1	0	0	1	1	1	1	0
	oldest								newest
Clock	0	1	0	1	0	1	0	1	0
Q3	1	1	1	0	0	1	1	1	1
Q2	1	1	1	1	1	0	0	1	1
Q1	1	1	1	1	1	1	1	0	0
Q0	1	1	1	1	1	1	1	1	1

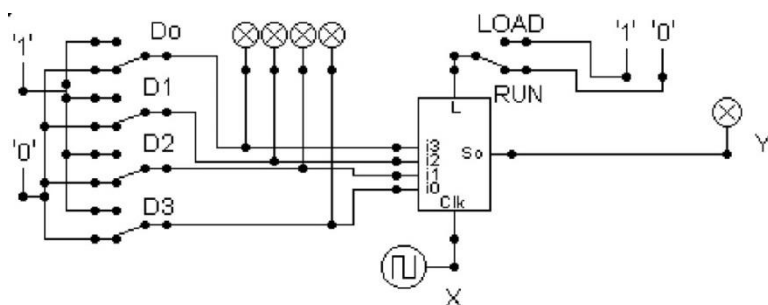
Describe what the circuit does. **(15 marks)**

it is noted that its from the range of oldest to the newest apparently it should differ. So when it's the oldest ,when the input is 1 and clock is 0 it the out come is Q3 ,Q2,Q1,Q0.=1.WHICH IS THE SAME OUT PUT FOR INPUT FOR BEING 1 AND CLOCK BEING 1 AND ALSO INPUT BEING 0 AND CLOCK BEING 0 , BUT DIFFERS WHEN THE INPUT IS 0 AND THE CLOCK IS ONE ,,,, THIS IS TOTALLY DIFFERENT WHEN IT COMES TO THE NEWEST ITS TOTALLY DIFFERENT

Combining everything we have covered so far (lectures 3-6).

Parallel to Serial converter

Open the LogSim circuit **week5.cct** from the Logsim folder. Also see week5 v2.cct for more help. It should look similar to this:



Describe what this circuit does. **(15 marks)**

~*“The circuit showcases how if the connection is correct as seen below when the input is 0110 it shows the out put is also 0110 this shows that even with the combination of the two circuit its possible to get the same output after the input”*

Design and add to the above circuit an additional circuit that takes the Clock X and the Output Y and decodes Y into 4 output indicators so that they match D0 – D3. Insert the LogSim GIF output of your design in the space below.

The highest marks will go to those who design the circuit such that it **AUTOMATICALLY** stops (not pauses) when the input to the circuit matches the output to the circuit

Note: Save your GIF image when your output indicators match the input D0 - D3. (30 marks)

